LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Demonstration of mechanism of typical instrument.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Demonstration of mechanism of typical instrument.
5. Specific Objectives	15.1.1.4 Trainee will be able to explain working of mechanisms in an instrument.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of instrument mechanism assembly, safety precautions regarding moving parts, instructor supervision during handling.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an aircraft instrument?
(b) Name two common types of aircraft instruments.
(c) Why is understanding the internal mechanism important for maintenance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to instrument types • Basic components of mechanism • Working principle of sensing element • Transmission of motion • Indicating system operation	L & Ds	PPT, IB	15 min
(b)	• Detailed mechanism disassembly • Function of gears and linkages • Role of damping mechanisms • Calibration and zeroing process • Common failure points analysis	L & Ds	PPT, PDF	20 min
(c)	• Reassembly of instrument mechanism • Verification of operational limits • Troubleshooting common faults • Maintenance and inspection procedures • Final functional check	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the sensing element converts physical quantity to mechanical motion.
(b) Describe the role of damping in ensuring accurate instrument readings.
(c) List three steps involved in the calibration of a typical instrument.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between the mechanisms of a magnetic compass and a gyroscopic instrument for the next lesson.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Quantitative displays: - Circular scale, high range long scale displays, angle of observation. Scale range and operating range. Straight scale digital display, dual indication and colored displays.	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min